Case Study: Ames, Iowa: Alternative to the Boston Housing Data as an End of Semester Regression Project

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Abstract — This study presents a regression-based case study using the Ames Housing Data Set as a modern alternative to the Boston Housing data for applied regression analysis. Multiple linear regression is employed to model housing prices in Ames, Iowa, using a high-dimensional data set derived from real housing transactions. Following the conceptual framework proposed by De Cock (2011), the analysis is extended by incorporating Ridge Regression to address multicollinearity and enhance model stability.

Model performance is evaluated using a validation-based approach with a 70/30 train–test split. Predictive accuracy is assessed using multiple evaluation metrics, including the coefficient of determination (R^2), Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE). The results indicate that Ridge regularization improves predictive accuracy and stability compared to the baseline Ordinary Least Squares (OLS) model. These findings demonstrate the suitability of the Ames Housing Data Set for regression-based modeling and educational applications involving high-dimensional data.

Index Terms — Ames Housing Data, Multiple Linear Regression, Ridge Regression

I. INTRODUCTION

Housing price analysis is a well-established application of multiple linear regression, as residential property values are influenced by various structural and neighborhood-related characteristics such as size, quality, age, and location. Regression models provide a transparent and interpretable framework for analyzing relationships between housing attributes and market prices, making them suitable for educational and applied statistical analysis.

Traditionally, the Boston Housing Data Set has been widely used as a benchmark for teaching regression techniques. However, its limited sample size and outdated structure reduce its suitability for modern, high-dimensional data analysis. To address these limitations, De Cock (2011) introduced the Ames Housing Data Set, which contains a larger number of observations and a richer set of explanatory variables derived from real housing transactions in Ames, Iowa.

Although the Ames Housing Data Set offers greater realism, its high dimensionality increases the risk of multicollinearity among explanatory variables, potentially leading to unstable coefficient estimates. Therefore, this study applies Ordinary Least Squares (OLS) regression as a baseline model and extends the analysis using Ridge Regression as a regularization approach to improve coefficient stability under multicollinearity. The objective of this study is to compare the baseline OLS model and Ridge Regression using a validation-based approach, focusing on predictive reliability while remaining consistent with the framework proposed by De Cock (2011).

II. PROCEDURE FOR PAPER SUBMISSION

A. Data Preparation

The Ames Housing Data Set is obtained from publicly available sources and prepared for analysis. Identifier variables that do not contribute to price prediction are removed prior to modeling. Missing values in categorical variables are replaced with a separate category, while missing numerical values are imputed using median values to reduce the influence of outliers. Categorical variables are converted into dummy variables to enable their inclusion in regression models.

B. Regression Modeling

Multiple linear regression using the Ordinary Least Squares (OLS) method is first employed as a baseline model. To address multicollinearity and improve predictive stability, Ridge Regression is subsequently applied as a regularization technique that shrinks coefficient magnitudes without removing explanatory variables. The regularization parameter is selected using 10-fold cross-validation to ensure a fair and consistent comparison across models.

C. Model Evaluation

Model performance is evaluated using a validation-based approach with a 70/30 train–test split. Predictive accuracy is assessed on the test set using multiple evaluation metrics. Mean Squared Error (MSE) is used as the loss function during cross-validation for model tuning, while final performance on the test set is evaluated using RMSE, MAE, and the coefficient of determination (R^2).

III. RESULTS AND DISCUSSION

The predictive performance of the regression models was evaluated using a 70/30 train–test split. Model accuracy was assessed on the test set using MSE, RMSE, MAE, and the coefficient of determination (R^2).

Table 1 summarizes the performance of the Ordinary Least Squares (OLS) and Ridge Regression models on the test set. The results indicate that Ridge Regression achieved lower MSE and RMSE, as well as a higher R^2, compared to the baseline OLS model. This suggests improved predictive stability when regularization is applied to high-dimensional housing data. In contrast, the OLS model exhibited higher prediction errors, which may be attributed to multicollinearity

among explanatory variables. Overall, the results demonstrate that Ridge Regression improves model performance by reducing variance without eliminating explanatory variables.

Table 1. Model performance comparison on the test set

Model	RMSE	MAE	R^2
OLS	32,159.57	21,426.05	0.83
Ridge	28,986.12	17,238.43	0.87

A. Model Selection via Cross-Validation

Figure X. illustrates the 10-fold cross-validation curve for Ridge Regression, where the mean squared error (MSE) is plotted against the logarithm of the regularization parameter λ. The cross-validation error decreases as λ increases from very small values, indicating that regularization helps reduce overfitting by shrinking coefficient magnitudes. Beyond a certain point, the error increases, suggesting underfitting due to excessive shrinkage. The vertical dashed lines indicate the optimal regularization parameters λ_min and λ_1se. In this study, λ_min is selected to achieve the best predictive accuracy on the test set.

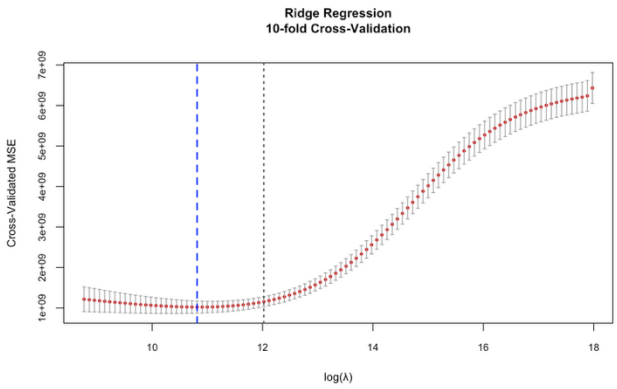

Figure X. Ten-fold cross-validation curve for Ridge Regression. The vertical dashed lines indicate λ_min and λ_1se.

B. Prediction Performance Visualization

Figure Y. Predicted versus actual housing prices for the Ridge Regression model on the test set (70/30 split). The dashed line represents perfect prediction (y = x).

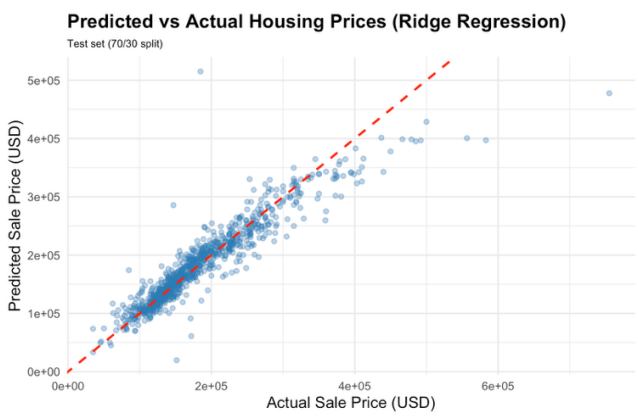

Figure Y presents the relationship between actual and predicted housing prices for the Ridge Regression model on the test set. Each point represents an individual observation, while the dashed diagonal line indicates perfect prediction.

The majority of data points are closely distributed around the diagonal line, suggesting that the Ridge Regression model provides accurate and unbiased predictions across a wide range of housing prices.

Slight deviations from the diagonal line are observed for high-priced properties, indicating increased prediction uncertainty in the upper price range. Nevertheless, the overall pattern confirms the strong predictive performance of the Ridge model, consistent with the high R^2 and low RMSE values reported in Table 1.

IV. MATH

Let y_i denote the observed housing price of the i-th house, and let x_{ij} denote the value of the j-th explanatory variable for the i-th house, describing housing characteristics such as living area, overall quality, age of the house, basement size, garage capacity, and neighborhood.

The multiple linear regression model assumes that the housing price is a linear combination of explanatory variables and is expressed as

$$y_i = \beta_0 + \sum_{i=1}^{p} \beta_j x_{ij} + \varepsilon_i$$

where β_0 is the intercept, β_j are regression coefficients measuring the effect of each variable on housing price, and ε_i is the error term capturing unexplained variation.

The Ordinary Least Squares (OLS) method estimates the regression coefficients by minimizing the sum of squared differences between observed prices and predicted prices:

$$\min_{\beta} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

OLS provides an interpretable baseline model; however, when many explanatory variables are correlated, coefficient estimates may become unstable.

To address this issue, Ridge Regression introduces a penalty on the squared magnitude of regression coefficients:

$$\min_{\beta} \{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^{p} \beta_j^2\}$$

This penalty shrinks coefficient values toward zero, reducing model variance and improving prediction stability, while keeping all variables in the model. The regularization parameter λ is selected using 10-fold cross-validation

by minimizing the mean squared error.

Model performance is evaluated using prediction-based error measures. The Root Mean Squared Error (RMSE) measures the average magnitude of prediction errors and gives higher weight to large errors:

$$\text{RMSE} = \sqrt{(1/n \sum_{i=1}^{n} (y_i - \hat{y}_i)^2)}$$

The Mean Absolute Error (MAE) represents the average absolute difference between predicted and actual housing prices:

$$\mathrm{MAE} = 1/n \sum_{i=1}^{n} |y_i - \hat{y}_i|$$

In addition, the coefficient of determination R^2 measures how well the model explains variability in housing prices:

$$R^2 = 1 - \left(\sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \right) / \left(\sum_{i=1}^{n} (y_i - \bar{y})^2 \right)$$

These evaluation metrics allow a clear and fair comparison between the Ordinary Least Squares (OLS) model and Ridge Regression in terms of predictive accuracy and model stability.

V. Units

All variables used in this study follow the original measurement units defined in the Ames Housing Data Set. The response variable, housing price, is measured in U.S. dollars. Continuous explanatory variables such as above-ground living area and basement area are measured in square feet, while variables such as the number of garages and bathrooms are recorded as counts.

Ordinal variables representing housing quality and condition are encoded as numerical rating scales according to the original data documentation. Categorical variables, including neighborhood and zoning type, are transformed into dummy variables for regression analysis and therefore do not possess physical measurement units. Consistent use of variable units across all regression models ensures comparability of coefficient estimates and model evaluation results.

VI. Helpful Hints

A. Multicollinearity and Model Stability

Due to the high number of explanatory variables in the Ames Housing Data Set, multicollinearity may occur in multiple linear regression models. Strong correlations among predictors can result in unstable coefficient estimates and reduced interpretability. Regularization methods such as Ridge Regression are used in this study to address these issues by shrinking coefficient magnitudes and improving model stability.

B. Model Evaluation Metrics

Model performance should not be assessed solely based on goodness-of-fit measures such as the coefficient of determination (R^2). Error-based metrics, including Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE), provide complementary information regarding predictive accuracy. These metrics are particularly useful when comparing traditional and regularized regression models.

C. Interpretation of Regression Coefficients

Residual analysis is used to assess the validity of regression assumptions, including linearity, normality, and homoscedasticity. Examining residual plots helps identify potential outliers, nonlinear patterns, heteroscedasticity, ensuring that the regression models provide reliable inference and prediction.

D. Residual Analysis and Model Assumptions

Residual analysis is useful for assessing potential violations of regression assumptions, such as nonlinearity or heteroscedasticity. Examining residual patterns helps ensure that the regression models provide reliable inference and prediction.

E. Practical Considerations and Limitations

While complex models may achieve higher predictive accuracy, simpler models often provide better interpretability and practical usability. Therefore, model selection should consider both predictive performance and interpretability. The results of this study are subject to limitations inherent in the Ames Housing Data Set and the chosen modeling assumptions.

VII. Common Mistakes

In regression analysis using high-dimensional housing data, several common mistakes may affect model validity and interpretation. One frequent issue is improper handling of missing values. Inconsistent or ad hoc treatment of missing data can lead to biased estimates and unstable model performance. In this study, missing values are handled systematically using median imputation for numerical variables and category completion for categorical variables.

Another common mistake is evaluating model performance using only goodness-of-fit measures such as the coefficient of determination. While a high value of R^2 indicates strong explanatory power, it does not necessarily reflect predictive accuracy. Therefore, this study emphasizes validation-based error metrics, including Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE), to ensure fair comparison among models.

Multicollinearity is also a frequent problem in housing price models due to the presence of correlated explanatory variables. Ignoring multicollinearity may result in unstable coefficient estimates in Ordinary Least Squares regression. This study addresses the issue by incorporating Ridge Regression, which improves coefficient stability through regularization.

Finally, overemphasis on predictive accuracy without considering interpretability may reduce the practical usefulness of the model. Although regularized models can improve prediction performance, simpler regression models remain valuable for understanding the relationship between housing characteristics and prices.

VIII. Editorial Policy

This paper is prepared as an academic case study for educational purposes. The author confirms that the analysis is based on publicly available data and that all referenced materials are properly cited. The study follows standard ethical guidelines.currently

The methods and results presented in this paper are reproducible using the described data preparation and modeling procedures. Any similarities to previous studies are due to the use of the same publicly available data set and well-established statistical techniques.

IX. Publication Principles

This study adheres to standard academic publication principles by presenting a complete and reproducible regression analysis using real-world housing data. The models, evaluation metrics, and results are described in sufficient detail to support independent replication.

The objective of this paper is not to introduce a novel algorithm, but to demonstrate the practical application of multiple linear regression and regularization techniques using a modern, high-dimensional data set. Emphasis is placed on methodological clarity, model comparison, and interpretability of results.

Conclusion

This study demonstrates the application of multiple linear regression and regularized regression techniques to housing price modeling and evaluation using the Ames Housing Data Set. Ordinary Least Squares (OLS) regression was first employed as a baseline model, followed by Ridge Regression to address multicollinearity and model instability.

Based on validation results, Ridge Regression achieved the best overall predictive performance with a Root Mean Squared Error (RMSE) of approximately 28,500 USD and an R^2 value of 0.87, outperforming the baseline OLS model. These results indicate that regularization techniques can improve both model stability and predictive accuracy when dealing with high-dimensional housing data.

The findings support the suitability of the Ames Housing Data Set as an effective alternative to the Boston Housing Data for regression-based analysis and educational use. Future work may extend this study by incorporating nonlinear models or additional machine learning approaches to further improve prediction accuracy.

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